

# Module 2: Agents — Think Like a Detective

## Learning Objectives

1. Develop the **mathematical detective mindset**: form hypotheses, test them with data, and update your thinking based on evidence.
2. Understand that a good problem-solver is an **active agent** who chooses strategies, not a passive rule-follower.
3. Learn to ask "**what would change my mind?**" — the most powerful question in mathematical reasoning.

## Introduction

Imagine a detective walks into a crime scene. Do they just stand there waiting for the answer to appear? Of course not. They observe the scene, form a theory about what happened, look for evidence that supports or contradicts their theory, and update their thinking as new evidence appears.

That's exactly what you should be doing with math problems. Too many students think math is about following rules: see this type of problem, apply this formula, get the answer. But real mathematical thinking is detective work. You observe the problem, form a hypothesis about how to solve it, try your approach, and update your strategy based on what happens.

The difference between a mediocre math student and a great one isn't about being "naturally smart." It's about being an active agent — someone who asks questions, tries strategies, and learns from what doesn't work.

# Key Concepts

## 1. Hypotheses: Your Best Guess (Before the Evidence)

A **hypothesis** is your best prediction about what's going on before you have all the information. Detectives form hypotheses. Scientists form hypotheses. And math detectives should too.

When you look at a word problem, before you start calculating, form a hypothesis: "I think the answer will be around 50." "I think this is a multiplication problem." "I think the pattern is doubling each time."

Why bother guessing before you solve it? Because your hypothesis gives you something to **check against**. If you predict the answer should be around 50 and you get 5,000, you know something went wrong. If you predicted doubling and the actual pattern is tripling, that prediction error tells you to revise your model.

Students who estimate before calculating catch more mistakes than students who just crunch numbers and hope for the best.

## 2. Strategy Selection: Choosing Your Approach

Being a mathematical agent means making **strategic choices** about how to approach problems, not just mechanically following procedures.

For any given problem, you might have multiple strategies: - **Draw a picture**: Great for geometry, spatial problems, and word problems with physical scenarios - **Make a table**: Great for finding patterns and organizing information - **Write an equation**: Great for problems with clear mathematical relationships - **Guess and check**: Great for getting started when you're stuck - **Work backwards**: Great for problems where you know the end result

An expert math student doesn't just know these strategies — they know *when* to use each one. And they're willing to switch strategies mid-problem if their first choice isn't working. That flexibility — trying something, noticing it's not working, and switching — is what makes someone an agent rather than just a calculator.

### 3. The Most Powerful Question: "What Would Change My Mind?"

Here's the question that separates good thinking from lazy thinking, in math and in life:

**"What would change my mind?"**

If you believe the next number in the sequence 2, 4, 8 is 16 (doubling), ask yourself: what evidence would prove me wrong? If the next number turned out to be 14 (adding increasing amounts: +2, +4, +6), you'd need to revise your model. Being willing to revise is what makes a good detective — and a good mathematician.

This question also helps you avoid **confirmation bias** — the tendency to only notice evidence that supports what you already believe. If you think a graph shows an upward trend, you might only focus on the points that go up and ignore the ones that go down. Asking "what would prove me wrong?" forces you to look at ALL the evidence, not just the convenient parts.

### Try It!

**Detective Notebook.** For your next math assignment, before you solve each problem, write down two things: (1) Your hypothesis — what you think the answer will be (or roughly what range), and (2) Your strategy — which approach you plan to use and why. After you solve the problem, compare your hypothesis to the actual answer and note whether your strategy was the right choice. Did you need to switch strategies? What would you do differently next time?

### Summary

Great mathematical thinking is **detective work**: form hypotheses, choose strategies, test your ideas, and update based on evidence. Being a mathematical **agent** means making active choices about how to approach problems, not just following procedures. The most powerful tool in your detective kit is the question **"what would change my mind?"** — it keeps you honest, flexible, and open to better answers. Next: **Perception** — how to spot real patterns in data (and avoid seeing patterns that aren't there).